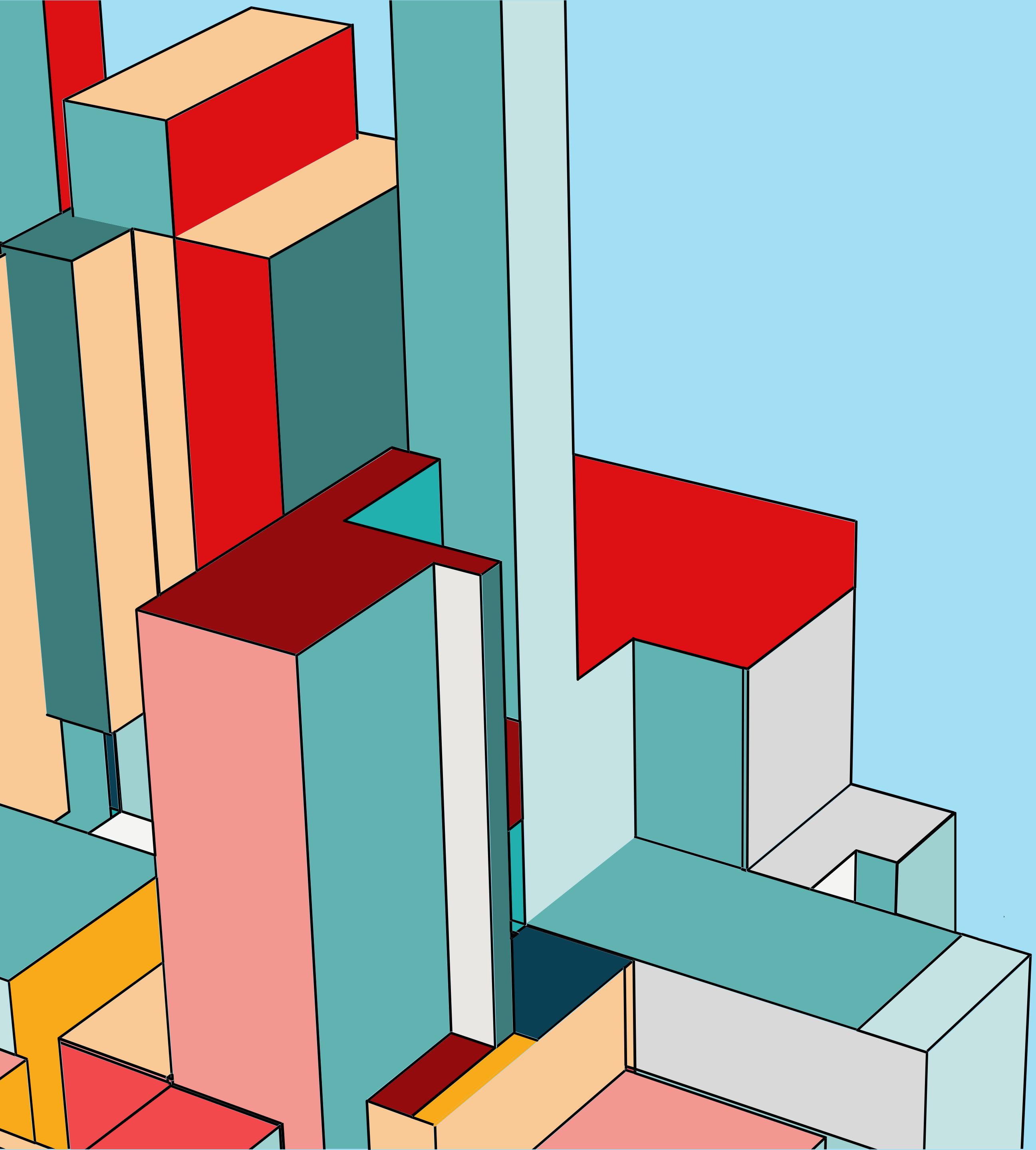
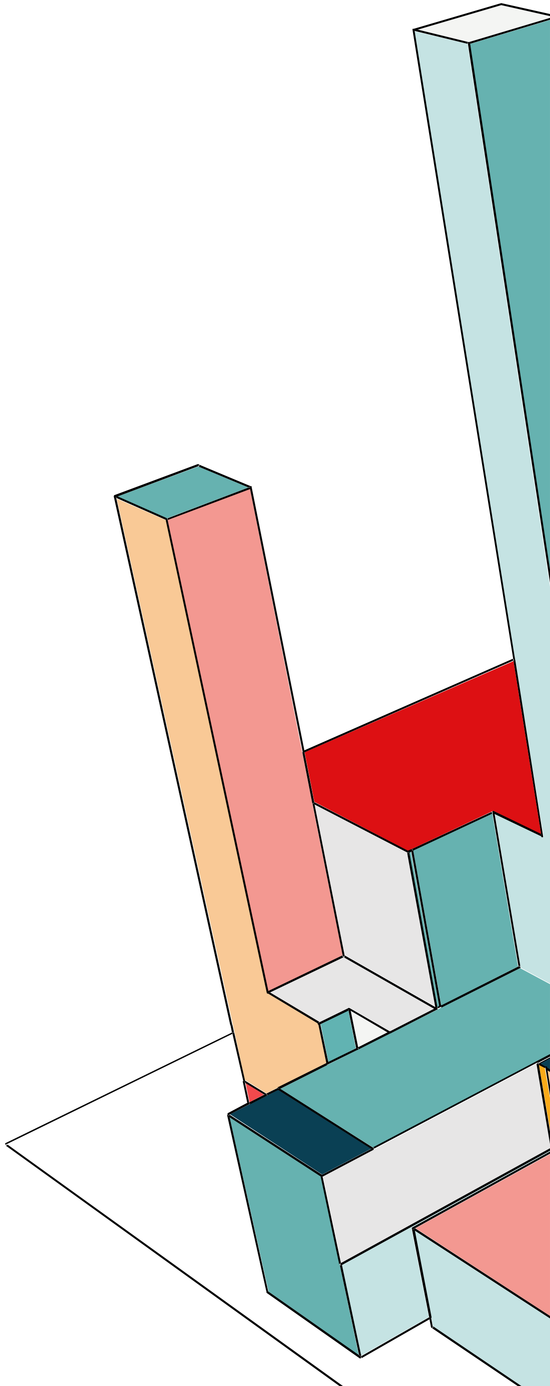


EGR-228 FINAL PROJECT

Cantilevered Beam

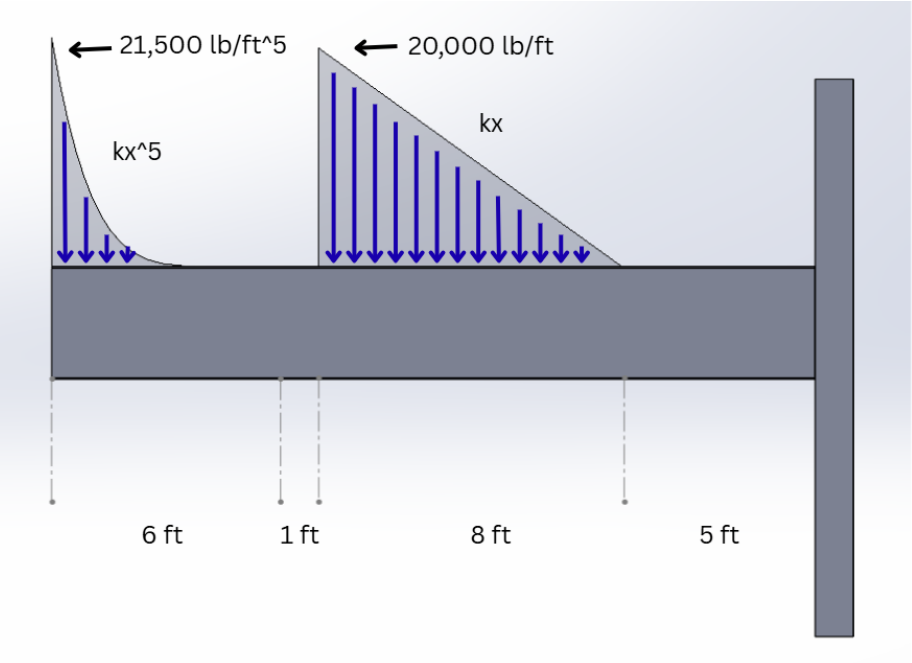
AGENDA

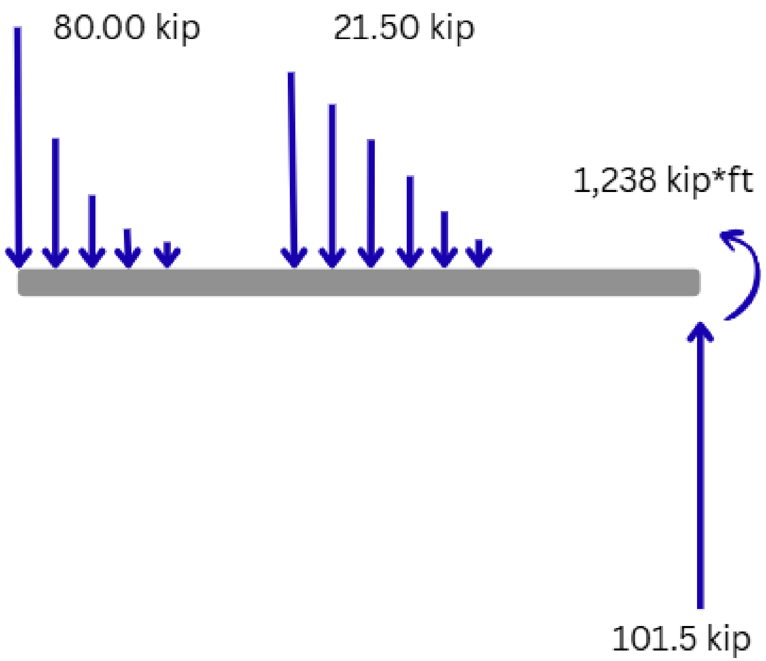
- Free Body Diagram
- Shear and Moment Diagram
- Solid Constant Beam
- Hollow Constant Beam
- Varied Beam
- Summary



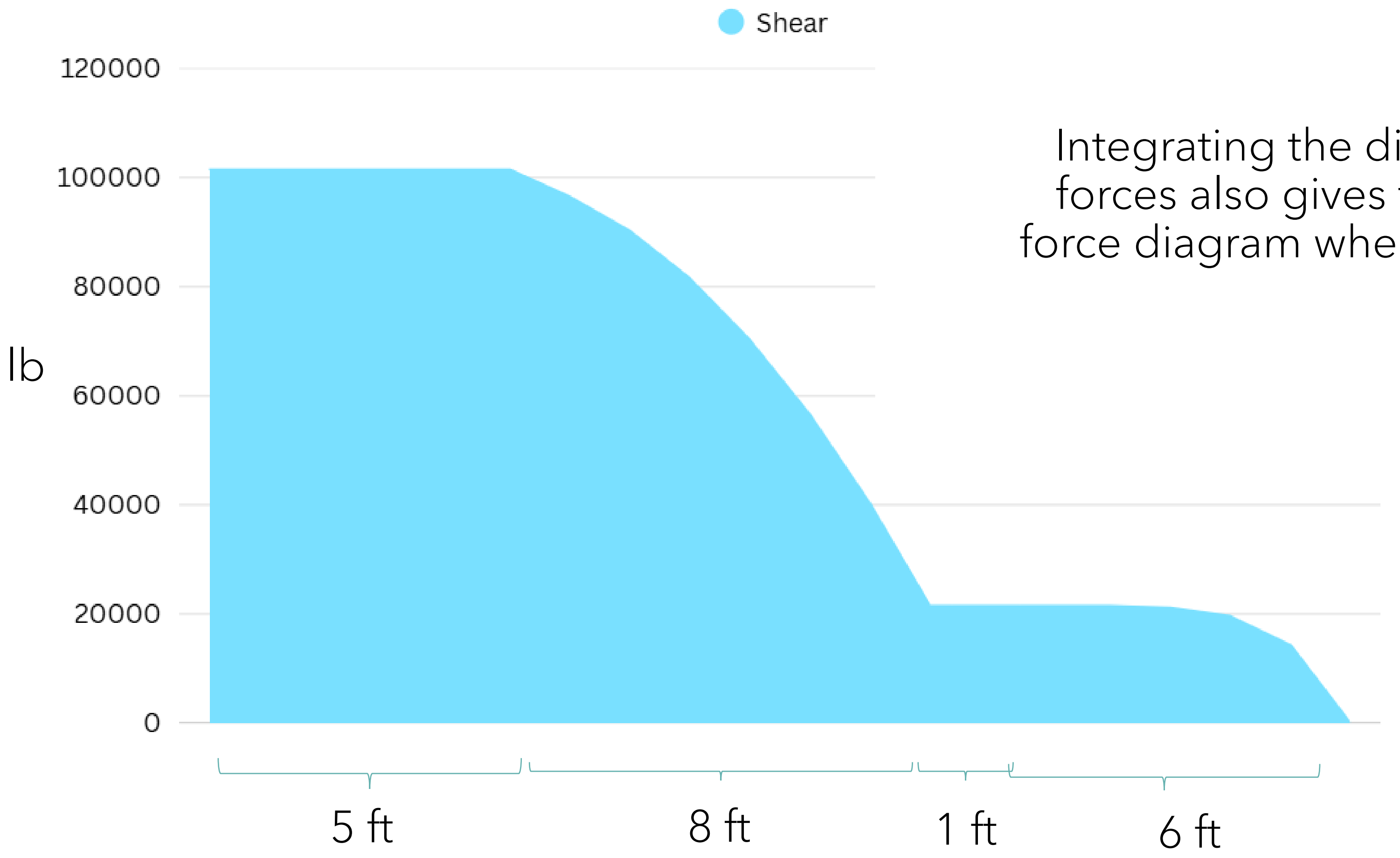
FREE BODY DIAGRAM

Integrating the distributed forces gives the total forces, which gives the ability to find the force and moment reactions:

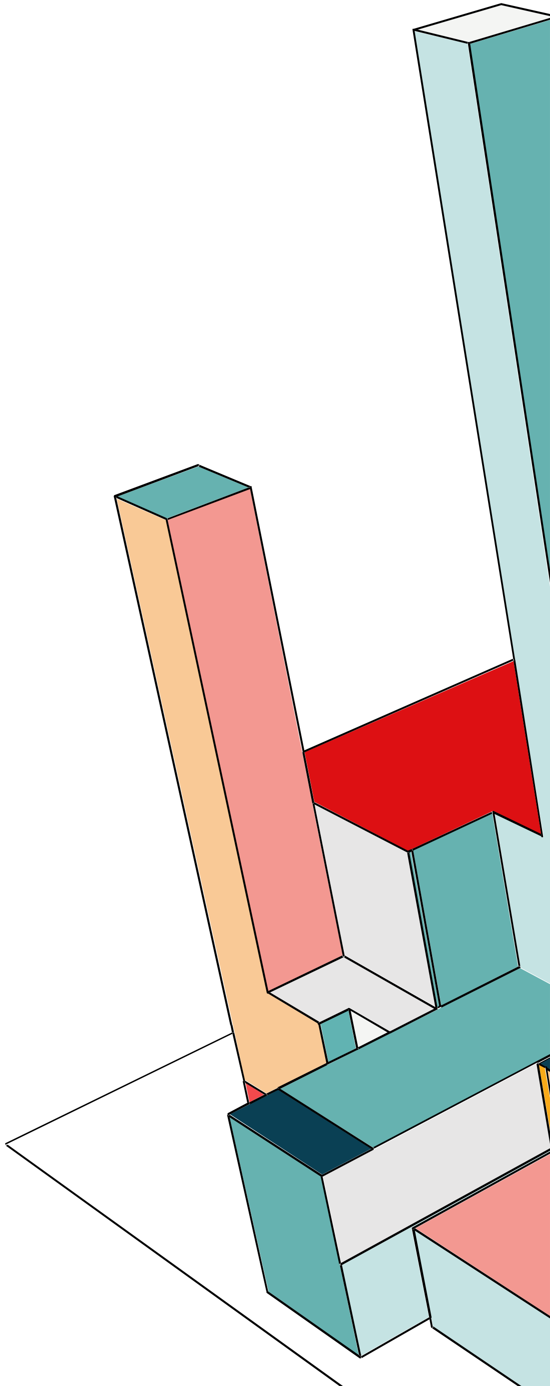




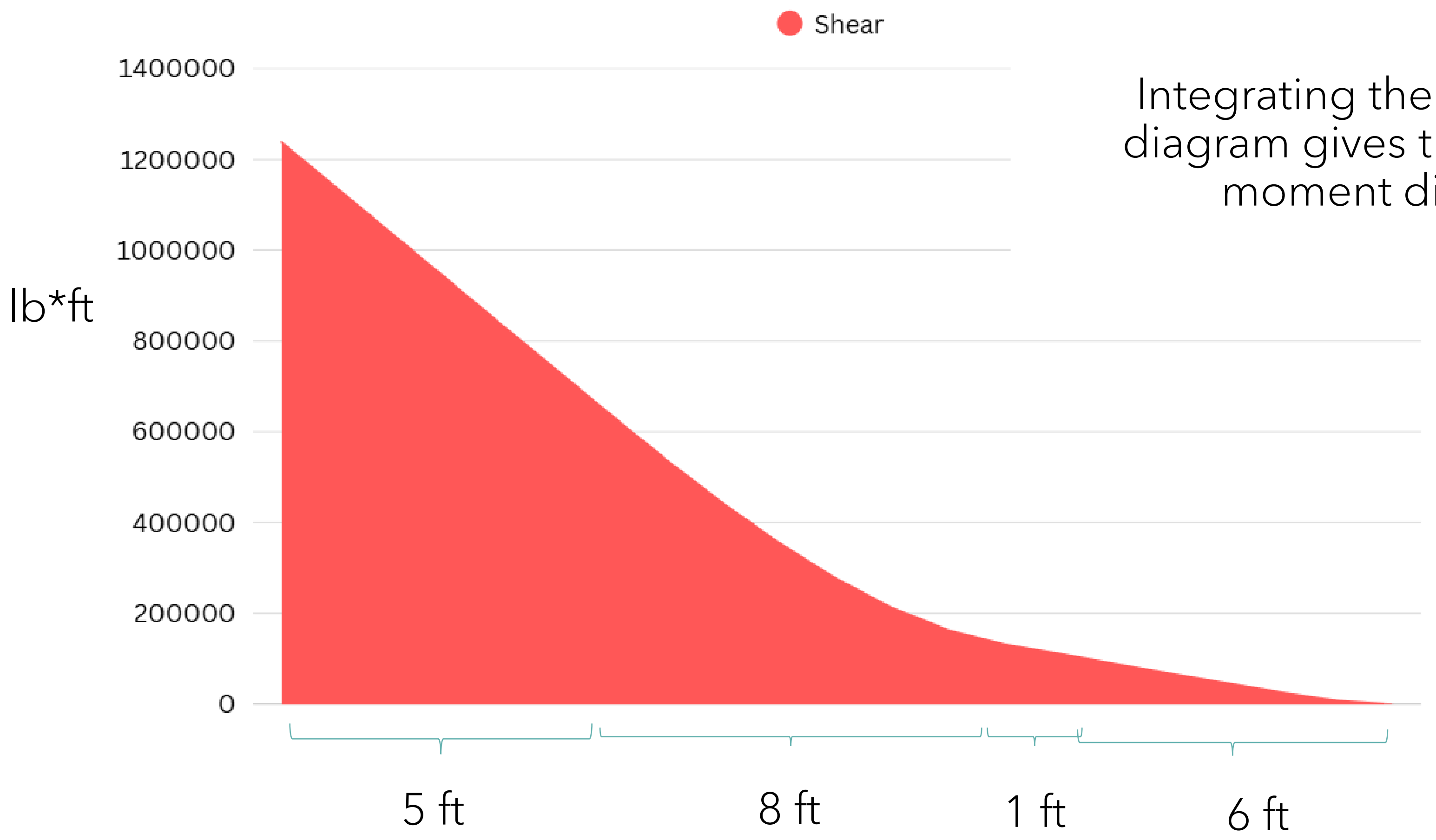
SHEAR DIAGRAM



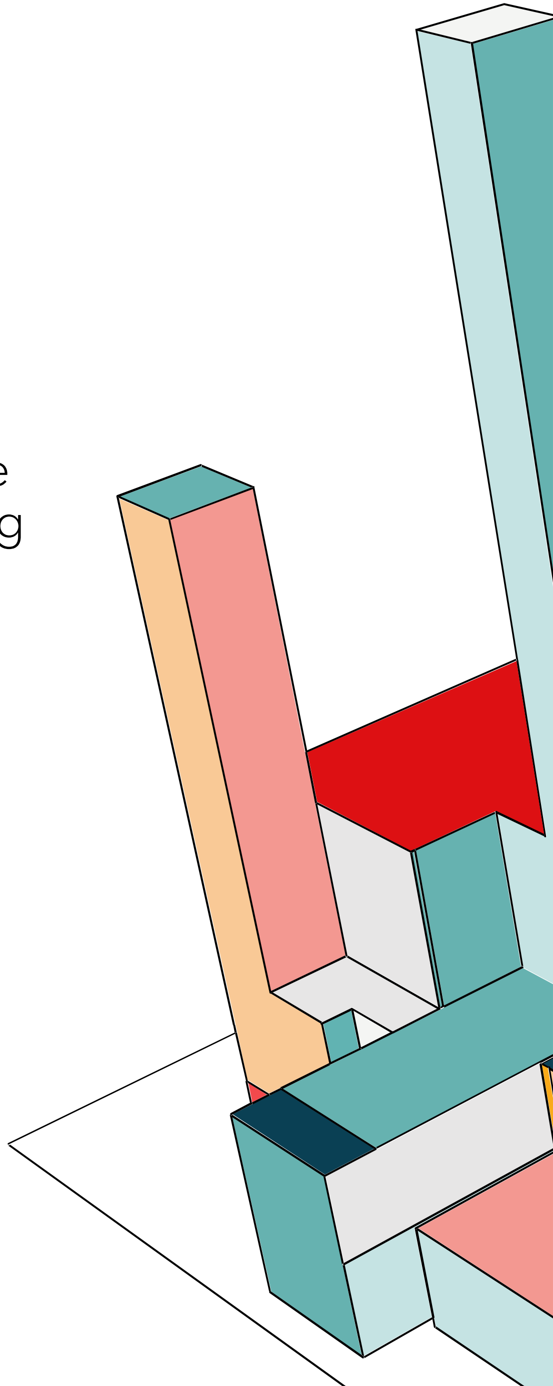
Integrating the distributed forces also gives the shear force diagram when graphed.

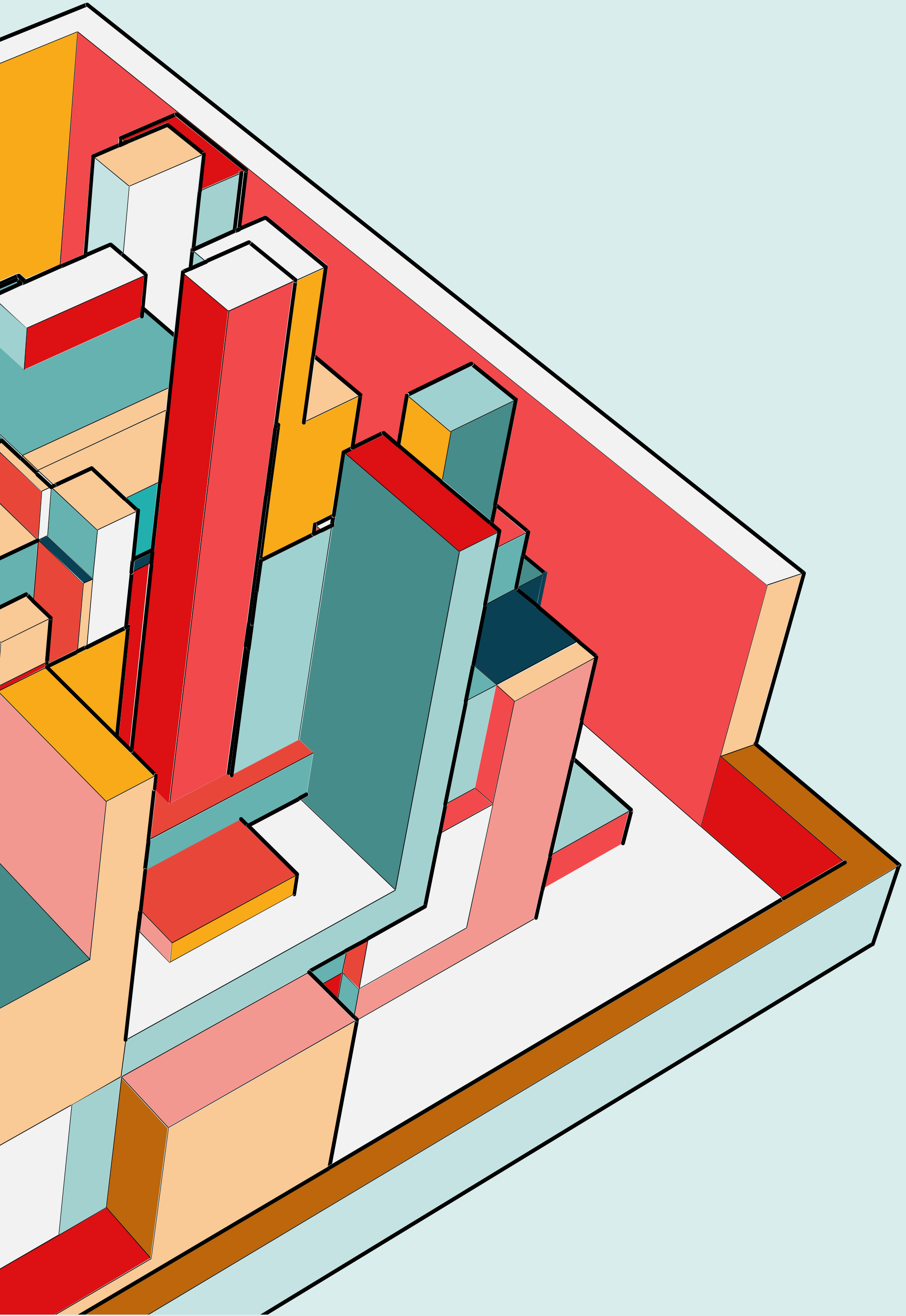


MOMENT DIAGRAM



Integrating the shear force diagram gives the following moment diagram.





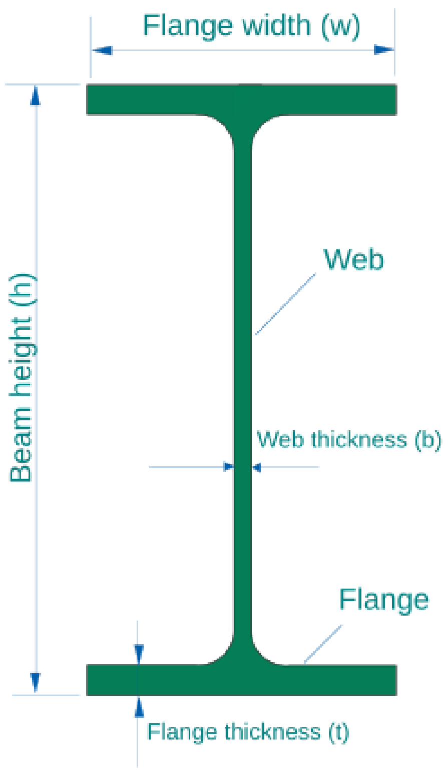
DESIGN CONSIDERATIONS

DESIGN REQUIREMENTS

The following beam will have to conform to the given requirements:

- A36 Steel Construction
- Factor of Safety = 4
- Minimum material thickness = ½ in
- Minimum beam width = 4 in
- Beam cross section should be symmetric across two axes.
- Given beam shape (solid constant, hollow constant, varied)
- Minimize weight

SOLID CONSTANT BEAM SHAPE



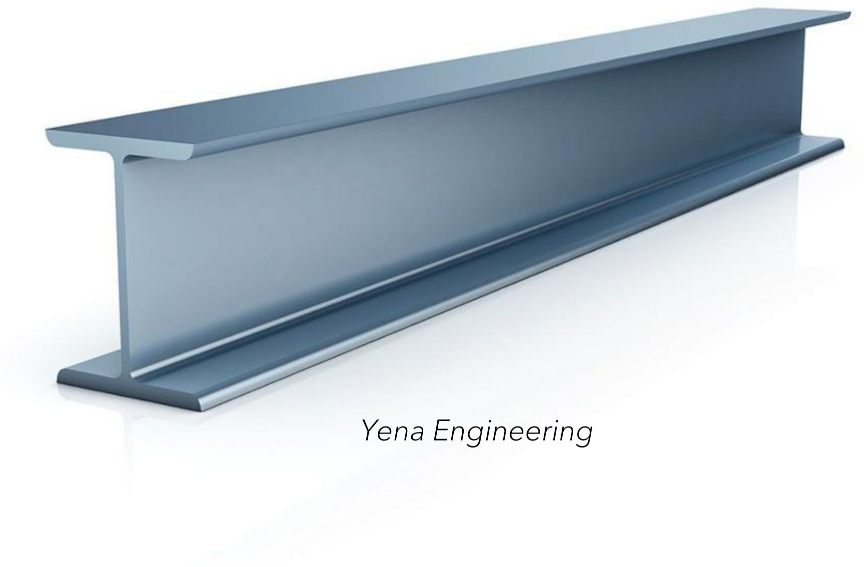
Wikipedia

An I-Beam shape was selected for its efficient resistance to shear and bending forces. An I-beam is comprised of two main components:

- The "web"
- The "flange"

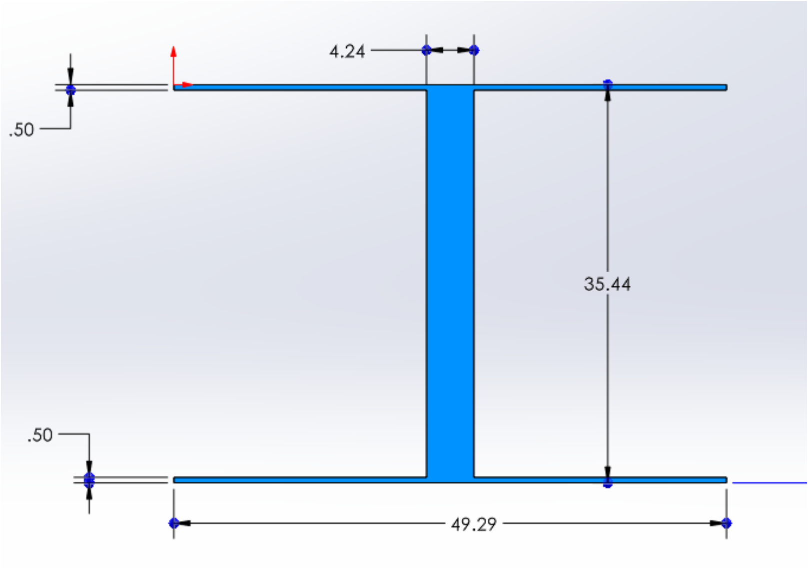
SOLID CONSTANT BEAM SHAPE

The "web", or middle portion of the I-beam can handle large shear forces, while the "flange" of the I-beam can handle large bending forces. This makes I-beams an attractive tool for supporting loads, particularly in civil engineering.



Yena Engineering

ULTIMATE SOLID-CONSTANT BEAM DESIGN

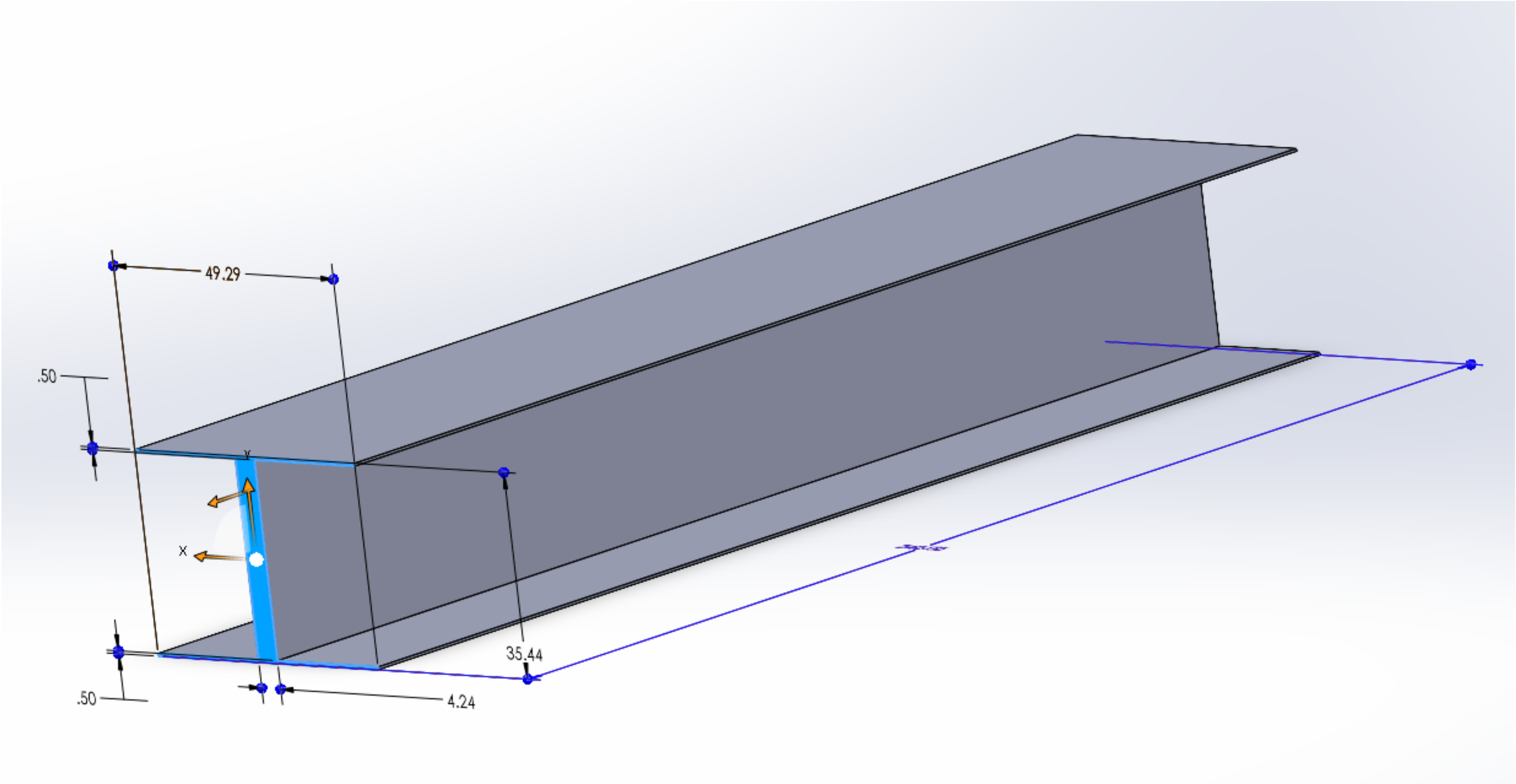


Height: 35.44 in

Base: 49.29 in

Flange thickness: 0.5000 in

Web thickness: 4.254 in



STRESS CALCULATIONS

$$\sigma_{allowed} = \frac{\sigma_{A36}}{F.S.} \qquad \tau_{allowed} = \frac{\tau_{A36}}{F.S.}$$

$$9\,000\ psi = \frac{36{,}000\ psi}{4} \qquad 5{,}000\ psi = \frac{20{,}000\ psi}{4}$$

The axial yield strength (36,000 psi) and shear yield strength of A36 steel (20,000 psi), as well as our designated factor of safety (4) are used to calculate the maximum stress allowed within the beam.

STRESS CALCULATIONS

The moment of inertia, section modulus, and beam area were all calculated using the following equations, respectively

$$I_x = \frac{bh^3 - \left(b - t_w\right)\left(h - 2t_f\right)^3}{12}$$

$$S_x = \frac{2I_x}{h}$$

$$A = bh - \left(\left(t_w\right)\left(h - 2t_f\right) + 2bt_f\right)$$

STRESS CALCULATIONS

$$\sigma_x = \frac{My}{I_x}$$

$$\tau_{xy} = \frac{VQ}{I_x t}$$

Normal stress and shear stress were calculated based off the maximum moments and shears, respectively, the beam faced. These occurred at the start of the beam, coinciding with the “wall” it was fixed to.

STRESS CALCULATIONS

Buckling was also considered as a potential source of failure, using a conservative k value of 1.0 (found in pin-pin supported beams). However, buckling was never found to be expected at any point in the beam and was not a determining factor for the shape of the beam.

$$P_{cr} = \frac{\pi^2 E I_x}{\left(k\left(h - 2t_f\right)\right)^2}$$

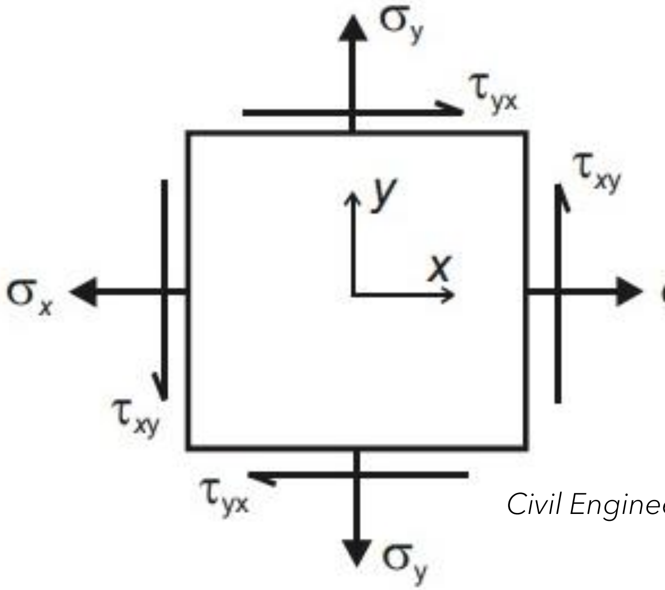
A k value of 1.0 is considered a conservative estimate here, because Euler's buckling equation would not factor the internal stresses of the flange or the stresses of the wall, which would both add rigidity to the beam and increase its critical load.

STRESS CALCULATIONS

$$\sigma_1 = \frac{\sigma_x}{2} + \sqrt{\left(\frac{\sigma_x}{2}\right)^2 + \tau_{xy}^2}$$

$$\sigma_2 = \frac{\sigma_x}{2} - \sqrt{\left(\frac{\sigma_x}{2}\right)^2 + \tau_{xy}^2}$$

$$\tau_{max} = \sqrt{\left(\frac{\sigma_x}{2}\right)^2 + \tau_{xy}^2}$$

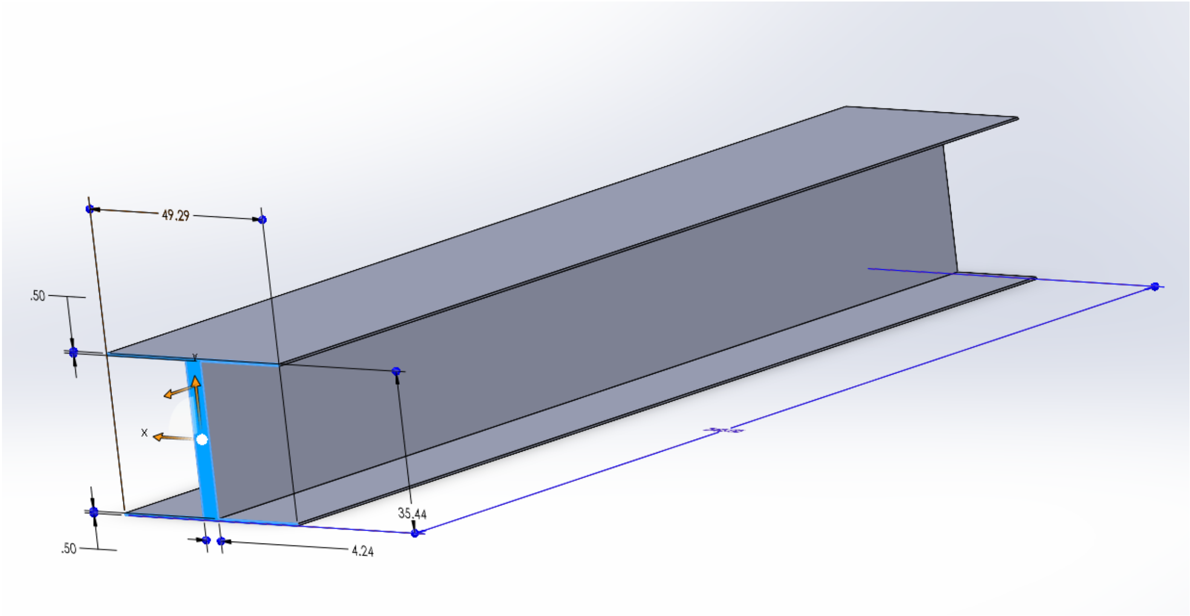


Civil Engineer Online

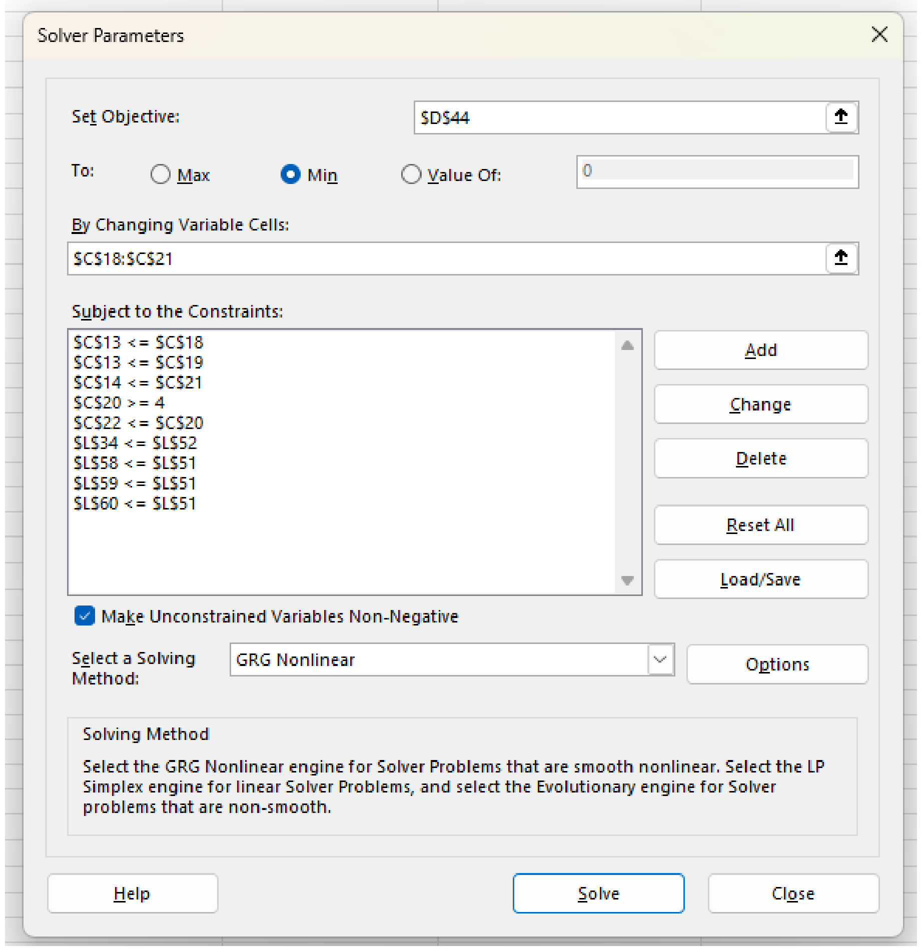
Lastly, principal stresses were determined using the following calculations. The stress applied to the beam would only be felt in one direction, so therefore, only one normal stress value was considered.

OPTIMIZATION

Each of the dimensions of the beam (height, width, flange thickness, web thickness) can be changed, and doing so will change both the area and strength of the beam. This means that this optimization problem cannot be solved with simple algebra and instead would require guessing and checking different dimensions.



OPTIMIZATION



Luckily, computational tools exist which can automate this task. Microsoft Excel contains an add on which can perform these solving operations, and is what was used to determine the dimensions of the beam.

By setting objective to minimize the area of the beam, setting constraints based on the constraints of the project, and allowing the program to change the four dimensions of the beam, an optimized beam shape was found.

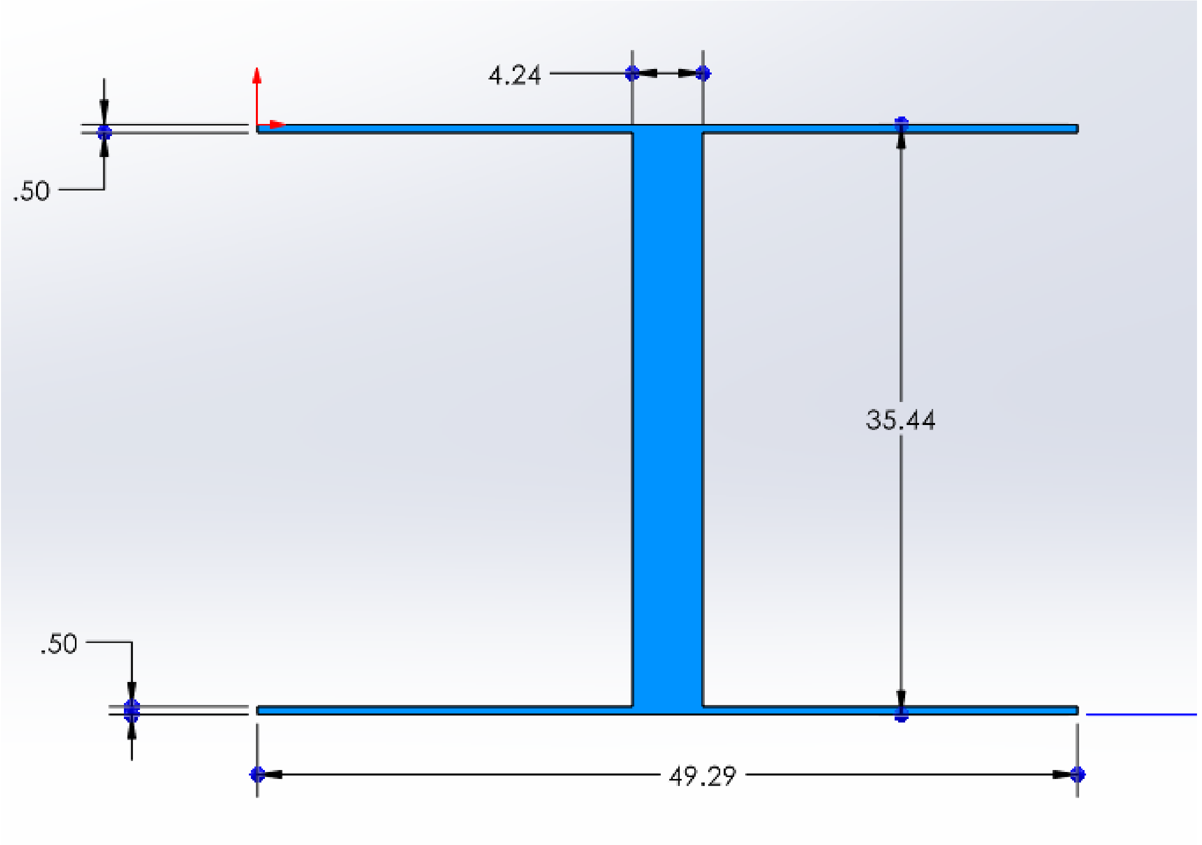
OPTIMIZATION

To properly automate this process, however, all proceeding necessary calculations had to also be automated. This was also done in excel, meaning the dimensions were able to be seamlessly modified while still returning accurate stress figures.

Formula	Variable	Calculation	Unit	Formula Used
b_5 = (-k_5 * g)	b_5 =	1,041.66667	lb/in	(-L5*C7)
b_4 = (-w_4 * (e + f + g))	b_4 =	0.00000	lb	0
k_5	k_5 =	-17.36111	lb/in^2	C10/C6
k_4	k_4 =	-9.25965E-07	lb/in^6	C11 / (C4 ^ 5)
F_5	F_5 =	-80,000.00000	lb	(L3 * C6) + (1 / 2) * (C6 * L5) * (C6 + (2 * C7))
F_4	F_4 =	-21,500.00000	lb	(1 / 6) * (L6) * (C5 + C6 + C7 - C3) ^ 6
F_1 = -(F_5 + F_6)	F_1 =	101,500.00000	lb	(-(L7+L8))
V_j = F_1	V_j =	101,500.00000	lb	L9
V_i = V_j + F_5	V_i =	21,500.00000	lb	L10 + L7
V_h = V_i + F_5	V_h =	21,500.00000	lb	L11
V_L = V_h + F_4	V_L =	0.00000	lb	L12 + L8
V_g = V_j	V_g =	101,500.00000	lb	L10
V_f = V_j + ((x * b_5) + (1 / 2) * x * k_5 * (x + (2 * g)))	V_f =	70,250.00000	lb	L10+(1/2)*(C24-C7)*(L5*(C24+C7)+(2*L3))
V_e = V_i	V_e =	21,500.00000	lb	L11
V_d = V_h + (1 / 6) * k_4 * (f + g + e - x) ^ 6	V_d =	-3,448,257.88752	lb	L12+(1/6)*(L6)*(C5+C6+C7-C24)^6
M_5 =	M_5 =	-9,920,000.00000	lb*in (CW)	(1/6)*(C6^2)*(L5)*((2*C6)+(3*C7))
M_4 =	M_4 =	-4,938,857.14286	lb*in (CW)	(1/42)*(L6)*((C6+C7-C3+C5)^6)*(C5+C6+C7+6*C3)
M_1 = -(M_5 + M_4)	M_1 =	14,858,857.14286	lb*in (CCW)	(-(L18+L19))
M_k = M_1	M_k =	14,858,857.14286	lb*in	L20
M_j = M_k - (V_j * g)	M_j =	8,768,857.14286	lb*in	L21 - (L10 * C7)
M_i =	M_i =	1,584,857.14286	lb*in	L22 - (1/6)*(C6)*(3*L3*C6+C6*L5*(C6+3*C7)+6*L10)
M_h = M_i - (V_h * e)	M_h =	1,326,857.14286	lb*in	L23 - (L12 * C5)
M_L =	M_L =	0.00000	lb*in	L24-(-L12*(C5+C6+C7-C3)-(1/42)*(L6)*(C5+C6+C7-C3)^7)
M_g = M_k - (V_j * x)	M_g =	14,858,857.14286	lb*in	L21 - (L10*C24)
M_f =	M_f =	14,233,857.14286	lb*in	L22 - (1/6)*(C24-C7)*(3*L3*(C24-C7)+(C24-C7)*L5*((C24-C7)+3*C7)+6*L10)
M_e = M_i - (V_i * (x-(f+g))	M_e =	4,938,857.14286	lb*in	L23-(L11*(C24-(C6+C7)))
M_d = M_h -	M_d =	-78,335,332.15755	lb*in	L24-(-L12*(C5+C6+C7-C24)-(1/42)*(L6)*(C5+C6+C7-C24)^7)

V_x = IFS(C24<C4,L16,C24<(C4+C5),L17,C24<(C4+C5+C6),L18,C24<=C3,L19)	V_x =	101,500.00000	lb	IFS(C24<=C7,L14,C24<=(C6+C7),L15,C24<=(C5+C6+C7),L16,C24<=C3,L17)
M_x = IFS(C24<C4,L20,C24<(C4+C5),L21,C24<(C4+C5+C6),L22,C24<=C3,L23)	M_x =	14,858,857.14286	lb * in	IFS(C24<=C7,L26,C24<=(C6+C7),L27,C24<=(C5+C6+C7),L28,C24<=C3,L29)
V_max_abs = ABS(V_max)	V_max_abs =	101,500.00000	lb	ABS(L32)
d_w = h - (2 * t_f)	d_w =	3.00000	in	C20 - (2 * C19)
A_f = b * t_f	A_f =	2.00000	in^2	C21 * C19
A_w = d_w * t_w	A_w =	4.09059	in^2	L35 * C18
A_total = (2 * A_f) + A_w	A_total =	8.09059	in^2	(2 * L36) + (L37)
I_w = (t_w * (d_w ^ 3)) / 12	I_w =	3.06794	in^4	(C18 * (L35 ^ 3)) / 12
I_f,local = (b * (t_f ^ 3)) / 12	I_f,local =	0.04167	in^4	(C21 * (C19 ^ 3)) / 12
y_f = (d_w + t_f) / 2	y_f =	1.75000	in	(L35 + C19) / 2
I_f = I_f,local + (A_f * (y_f ^ 2))	I_f =	6.16667	in^4	L40 + (L36 * (L41 ^2))
I = I_w + (2 * I_f)	I =	15.40127	in^4	L39 + (2 * L42)
c = h / 2	c =	2.00000	in	C20 / 2
σ_x = (M_max * c) / I	σ_x =	-1,929,562.24915	psi	(-L33 * L44) / L43
Q = (A_f * y_f + A_thw * y_w)	Q =	5.03397	in^3	(L36 * L41) + (L47 * L48)
A_thw = (t_w * d_w) / 2	A_thw =	2.04529	in^2	(C18 * L35) / 2
y_w = d_w / 4	y_w =	0.75000	in	L35 / 4
τ_xy = (V_max * Q) / (I * t_w)	τ_xy =	-24,330.76590	psi	(-L32 * L46) / (L43 * C18)
F_max_buckle = ((π ^ 2) * E * I_x) / ((d_w * k) ^ 2)	V_max_buckle =	13,374.05005	lbs	((PI() ^ 2) * C17 * L53) / ((L3 * C16) ^ 2)
σ_all = σ_y / F.S.	σ_all =	9,000.00000	psi	C15 / C12
F_all = F_max_buckle / F.S.	F_all =	3,343.51251	lbs	L50 / C12
I_x = (L_beam * (t_w ^ 3)) / 12	I_x =	50.70172	in^4	(C3 * (C18 ^ 3)) / 12
S = I/c	S =	7.70064	in^3	L43/L44
σ_max = (σ_x / 2) + ((σ_x / 2) ^ 2 + (τ_xy ^ 2)) ^ 0.5	σ_max =	306.74941	psi	ABS((L45 / 2) + ((L45 / 2) ^ 2 + (L49 ^ 2)) ^ 0.5)
σ_min = (σ_x / 2) - ((σ_x / 2) ^ 2 + (τ_xy ^ 2)) ^ 0.5	σ_min =	1,929,868.99856	psi	ABS((L45 / 2) - ((L45 / 2) ^ 2 + (L49 ^ 2)) ^ 0.5)
τ_max = ((σ_x / 2) ^ 2 + (τ_xy ^ 2)) ^ 0.5	τ_max =	965,087.87398	psi	ABS(((L45/2)^2+(L49^2))^0.5)

SOLID-CONSTANT BEAM



Thus, the following dimensions were found for the solid constant beam:

Height: 35.44 in

Base: 49.29 in

Flange Thickness: 0.5000 in

Web Thickness: 4.254 in

Area: 195.8 in^2

Section Modulus: 15,330 in^3

Moment of Inertia: 271,600 in^4

Volume: 46,990 in^3

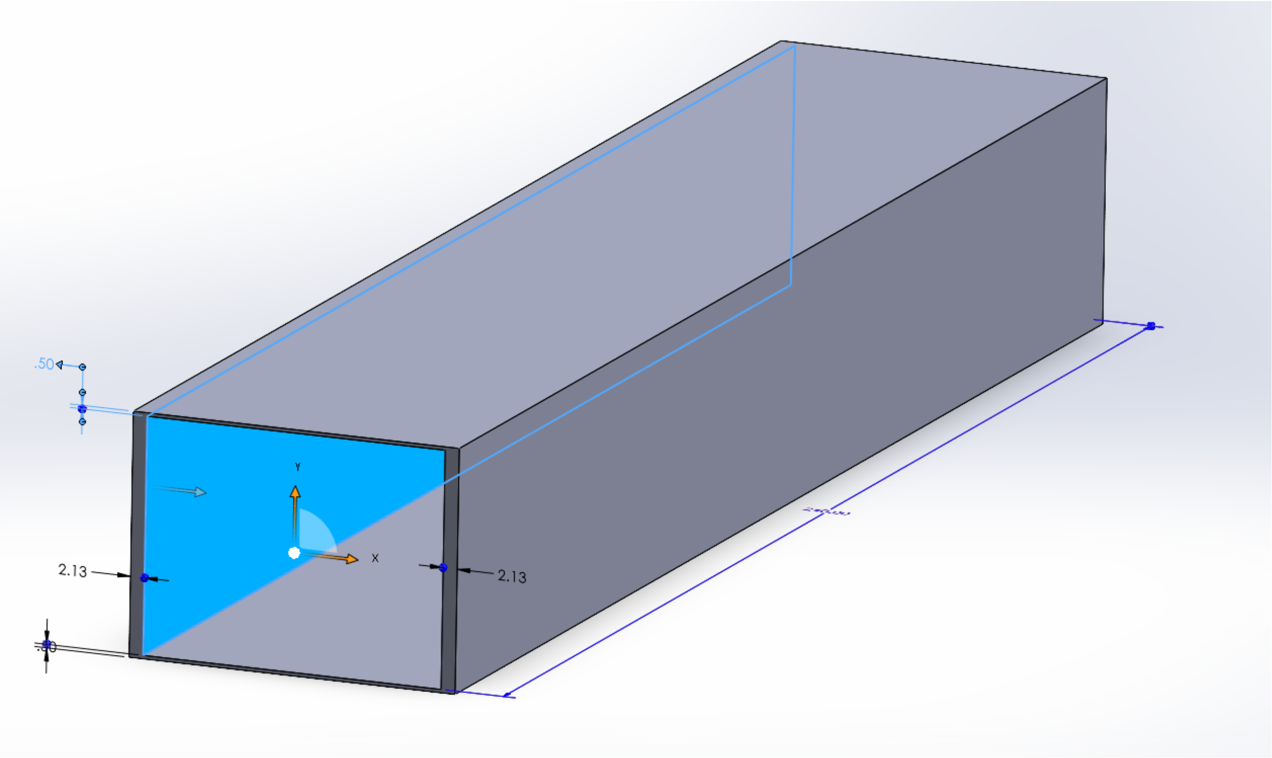
Weight: 13,340 lb

HOLLOW CONSTANT BEAM

These simulations were unable to find a hollow constant design that was stronger than the solid constant beam already derived. If the beam's web were to be split in two across the middle to allow for a "gap", the normal and shear stress characteristics would stay largely the same throughout the beam, however the risk for buckling would only increase due to the thinner supporting members.

However, a hollow beam would be superior dealing with forces applied from all directions, such as those applied perpendicular to the forces applied to this beam.

HOLLOW-CONSTANT BEAM



Thus, the following dimensions were found for the solid constant beam:

Height: 35.44 in

Base: 49.29 in

Flange Thickness: 0.5000 in

Side Wall Thickness: 2.127 in

Area: 195.8 in^2

Section Modulus: 15,330 in^3

Moment of Inertia: 271,600 in^4

Volume: 46,990 in^3

Weight: 13,340 lb

VARIED BEAM

While the Excel solver tool makes the process of finding the optimal shape of the beam much quicker, it would be impractical for one to do so for every 1-inch increment throughout the length of the beam. Therefore, a certain process was used where the properties of the beam could be approximated as a function of x.

x =	60	1.363528583	0.5	4	4	8.09058575	9.234605979
	Section #	Section Length (in)	Section Start (in)	Section End (in)	Step (in)		
	1	60	0	60	6		

Section 1 (x) (in)	t_w (in)	t_f (in)	h (in)	b (in)	A (in²)	S (in³)
0	4.253554	0.500000	35.438795	49.293334	195.780608	15,327.232783
6	4.253554	0.500000	34.763528	48.175235	191.790224	14,456.648874
12	4.253554	0.500000	34.047363	47.145571	187.714310	13,572.505883
18	4.253554	0.500000	33.313595	46.099559	183.547177	12,704.923468
24	4.253554	0.500000	32.565404	45.017248	179.282397	11,858.924525
30	4.253554	0.500000	31.796853	43.916715	174.912792	11,030.039275
36	4.253554	0.500000	31.013467	42.766248	170.430146	10,225.401828
42	4.253554	0.500000	30.192114	41.654951	165.825184	9,426.172373
48	4.253554	0.500000	29.361049	40.452189	161.087440	8,659.934159
54	4.253554	0.500000	28.489299	39.277521	156.204738	7,903.202656
60	4.253554	0.500000	27.638919	37.852578	151.162655	7,205.108770

Function	Form	a	b	c	d	R^2
t_w(x) =	ax^3 + bx^2 + cx + d	0.0000000000	0.0000000000	0.0000000000	4.2535539235	0.0161616162
t_f(x) =	ax^3 + bx^2 + cx + d	0.0000000000	0.0000000000	0.0000000000	0.5000000000	1.0000000000
h(x) =	ax^3 + bx^2 + cx + d	0.0000002420	-0.0003111638	-0.1123662811	35.4414397728	0.9986078830
b(x) =	ax^3 + bx^2 + cx + d	-0.0000079517	0.0003453381	-0.1820568436	49.2853769559	0.9986078830
A(x) =	ax^3 + bx^2 + cx + d	-0.0000069221	-0.0009782140	-0.6600128793	195.7838982324	1.0000000000
S(x) =	ax^3 + bx^2 + cx + d	0.0018113181	0.0981147371	-147.9277494359	15331.1696718040	0.9986871241

	Form	Average Area (in²)	Volume (in³)	Weight (lb)		
	ax^3 + bx^2 + cx + d	174.4358605470	10466.1516328228	2972.3870637217		

VARIED BEAM

This process involves the following steps:

1. Split the beam into 5 sections.
2. Split each section into 10 “steps”, based on x position.
3. For each step in each section, run the excel solver to find the optimal shape for that given portion of the beam.
4. Once the dimensions for each of these steps is found, use Excel’s cubic regression function (=LINEST), to calculate an approximate formula for each dimension.
5. Using Excel’s R^2 calculator (=RSQ), gauge the accuracy of the formula.

VARIED BEAM

This process involves the following steps:

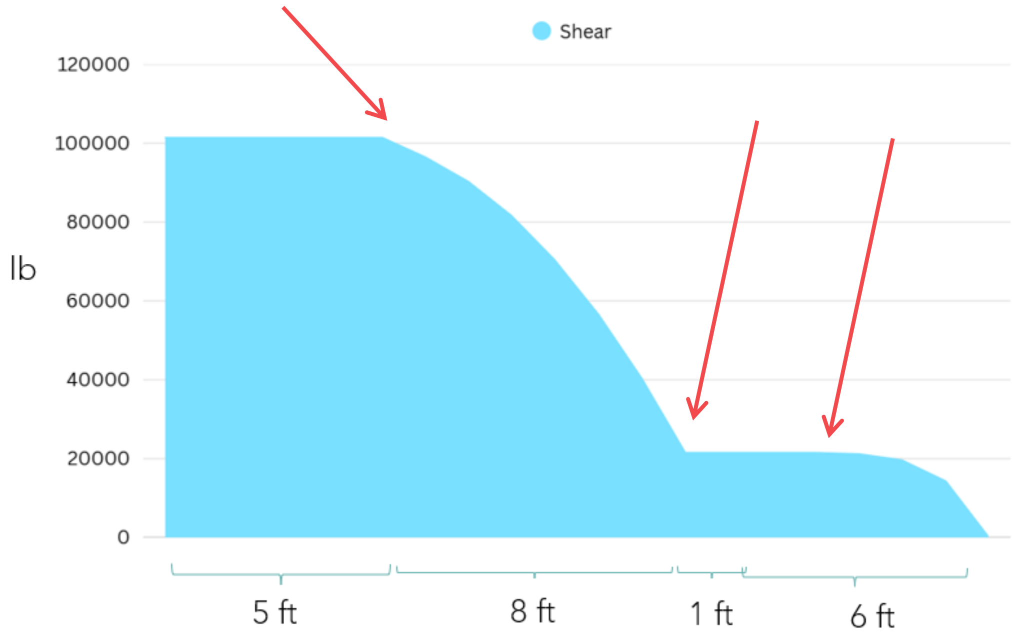
6. Using the equations for each dimension, write 5 new equations approximating the area of the beam as a function of x.
7. Integrate the equations of the area to find the equations for the total volume of the varied beam.
8. Using these equations, automation can be used to approximate every x value with any size step (x=1in was used in the spreadsheet).

VARIED BEAM

There are two reasons why sections are used, rather than splitting the entire beam into 10 steps and integrating from there.

VARIED BEAM

1. As x changes throughout the beam, the load on the beam changes irregularly. This means that there will be sudden fluctuations in the optimal dimensions for the beam, rather than smooth and formulaic shear and moment trends. Each section can therefore be started at these cusps, meaning each section will have its own equation for the dimensions of its beam.



VARIED BEAM

2. As the dimensions change across the beam, irregular changes in the patterns of optimal beam dimensions can occur, depending largely on the constraints of the beam. For example, as the optimal width of the beam decreases to 4 inches, the minimum allowed width of the beam, the width will suddenly stop decreasing as x increases. Instead, the solver will look for other ways to reduce area, such as reducing more height or flange/web thickness. This leads to cusps in the data, which will negatively affect the accuracy of the final equations. Therefore, sections can be added at these points to retain the accuracy of the equations.

VARIED BEAM

Section	x (in)
1	0 - 60
2	60 - 156
3	156 - 168
4	168 - 198
5	198 - 240

VARIED BEAM (SECTION 1, SECTION 2)

Section 1	Form	a	b	c	d	R^2
t_w(x) =	ax^3 + bx^2 + cx + d	0.0000000000	0.0000000000	0.0000000000	4.2535539235	1.0000000000
t_f(x) =	ax^3 + bx^2 + cx + d	0.0000000000	0.0000000000	0.0000000000	0.5000000000	1.0000000000
h(x) =	ax^3 + bx^2 + cx + d	0.0000002420	-0.0003111638	-0.1123662811	35.4414397728	0.9986078830
b(x) =	ax^3 + bx^2 + cx + d	-0.0000079517	0.0003453381	-0.1820568436	49.2853769559	0.9986078830
A(x) =	ax^3 + bx^2 + cx + d	-0.0000069221	-0.0009782140	-0.6600128793	195.7838982324	1.0000000000
S(x) =	ax^3 + bx^2 + cx + d	0.0018113181	0.0981147371	-147.9277494359	15331.1696718040	0.9986871241

	Form	Average Area (in^2)	Volume (in^3)	Weight (lb)		
	ax^3 + bx^2 + cx + d	174.4358605470	10466.1516328228	2972.3870637217		

Section 2	Form	a	b	c	d	R^2
t_w(x) =	ax^3 + bx^2 + cx + d	-0.0000021159	0.0000639647	-0.0044042939	4.2681103364	0.8664793790
t_f(x) =	ax^3 + bx^2 + cx + d	0.0000000000	0.0000000000	0.0000000000	0.5000000000	1.0000000000
h(x) =	ax^3 + bx^2 + cx + d	0.0000104421	-0.0010178319	-0.1290077037	27.5732097007	0.9880114255
b(x) =	ax^3 + bx^2 + cx + d	0.0000065714	-0.0010400642	-0.2221574835	37.8869919083	0.9880114255
A(x) =	ax^3 + bx^2 + cx + d	0.0000294065	-0.0049807670	-0.8592152310	151.2154770125	0.9995652113
S(x) =	ax^3 + bx^2 + cx + d	0.0003494543	0.5332360219	-122.2409177831	7214.3940469997	0.9993004550

	Form	Average Area (in^2)	Volume (in^3)	Weight (lb)		
	ax^3 + bx^2 + cx + d	101.1744826315	9712.7503326249	2758.4210944655		

VARIED BEAM (SECTION 3, SECTION 4)

Section 3	Form	a	b	c	d	R^2
t_w(x) =	ax^3 + bx^2 + cx + d	0.0000000000	0.0000000000	0.0000000000	2.5355727870	1.0000000000
t_f(x) =	ax^3 + bx^2 + cx + d	0.0000000000	0.0000000000	0.0000000000	0.5000000000	1.0000000000
h(x) =	ax^3 + bx^2 + cx + d	0.0012684499	-0.0277583607	0.0756359162	15.1951966432	0.9114071834
b(x) =	ax^3 + bx^2 + cx + d	0.0010704458	-0.0239870367	0.0680317009	12.7433206992	0.9114071834
A(x) =	ax^3 + bx^2 + cx + d	0.0042866929	-0.0943703807	0.2598120718	48.7362750143	0.9044353571
S(x) =	ax^3 + bx^2 + cx + d	0.1799692515	-3.8459145770	10.3730849228	695.2761490062	0.9100514340

	Form	Average Area (in^2)	Volume (in^3)	Weight (lb)		
	ax^3 + bx^2 + cx + d	47.6172205240	571.4066462878	162.2794875457		

Section 4	Form	a	b	c	d	R^2
t_w(x) =	ax^3 + bx^2 + cx + d	-0.0000005666	0.0000159709	-0.0001157989	2.5356894889	0.6022670012
t_f(x) =	ax^3 + bx^2 + cx + d	0.0000000000	0.0000000000	0.0000000000	0.5000000000	1.0000000000
h(x) =	ax^3 + bx^2 + cx + d	-0.0000083577	-0.0002141599	-0.1097672582	14.2764029270	0.9983501752
b(x) =	ax^3 + bx^2 + cx + d	-0.0000008244	-0.0007034606	-0.0840958124	11.9200248871	0.9983501752
A(x) =	ax^3 + bx^2 + cx + d	-0.0000271013	-0.0011082662	-0.3633944991	45.5842671933	0.9964204795
S(x) =	ax^3 + bx^2 + cx + d	0.0000004572	0.0599485238	-13.3690468558	573.6134034528	0.9978216336

	Form	Average Area (in^2)	Volume (in^3)	Weight (lb)		
	ax^3 + bx^2 + cx + d	39.6178664038	1188.5359921130	337.5442217601		

VARIED BEAM (SECTION 5)

Section 5	Form	a	b	c	d	R^2
t_w(x) =	ax^3 + bx^2 + cx + d	-0.0000246001	0.0005564746	-0.0056410108	2.5324771395	0.7761063430
t_f(x) =	ax^3 + bx^2 + cx + d	0.0000000000	0.0000000000	0.0000000000	0.5000000000	1.0000000000
h(x) =	ax^3 + bx^2 + cx + d	0.0000532098	-0.0039716318	-0.1159512045	10.5442755660	0.9939932645
b(x) =	ax^3 + bx^2 + cx + d	0.0001686088	-0.0077125716	-0.0742859543	8.7462161473	0.9939932645
A(x) =	ax^3 + bx^2 + cx + d	0.0002904056	-0.0184237382	-0.3776723151	32.9058085080	0.9880518458
S(x) =	ax^3 + bx^2 + cx + d	0.0018344682	0.0271605220	-9.3749550149	225.9597678421	0.9950354002

Form	Average Area (in^2)	Volume (in^3)	Weight (lb)		
ax^3 + bx^2 + cx + d	19.5676855453	821.8427929040	233.4033531847		

VARIED BEAM SUMMARY

The varied beam design greatly saved on material costs compared to the constant beams, due to being able to save weight where it was not needed. In fact, the varied beam used 48.44% of the steel the constant beam did, thus being 106.4% more efficient with its steel.

Varied Beam:

Average area: 94.84 in^2

Volume: 22,760 in^3

Weight: 6,464 lb

Constant Beam:

Average area: 195.8 in^2

Volume: 46,990 in^3

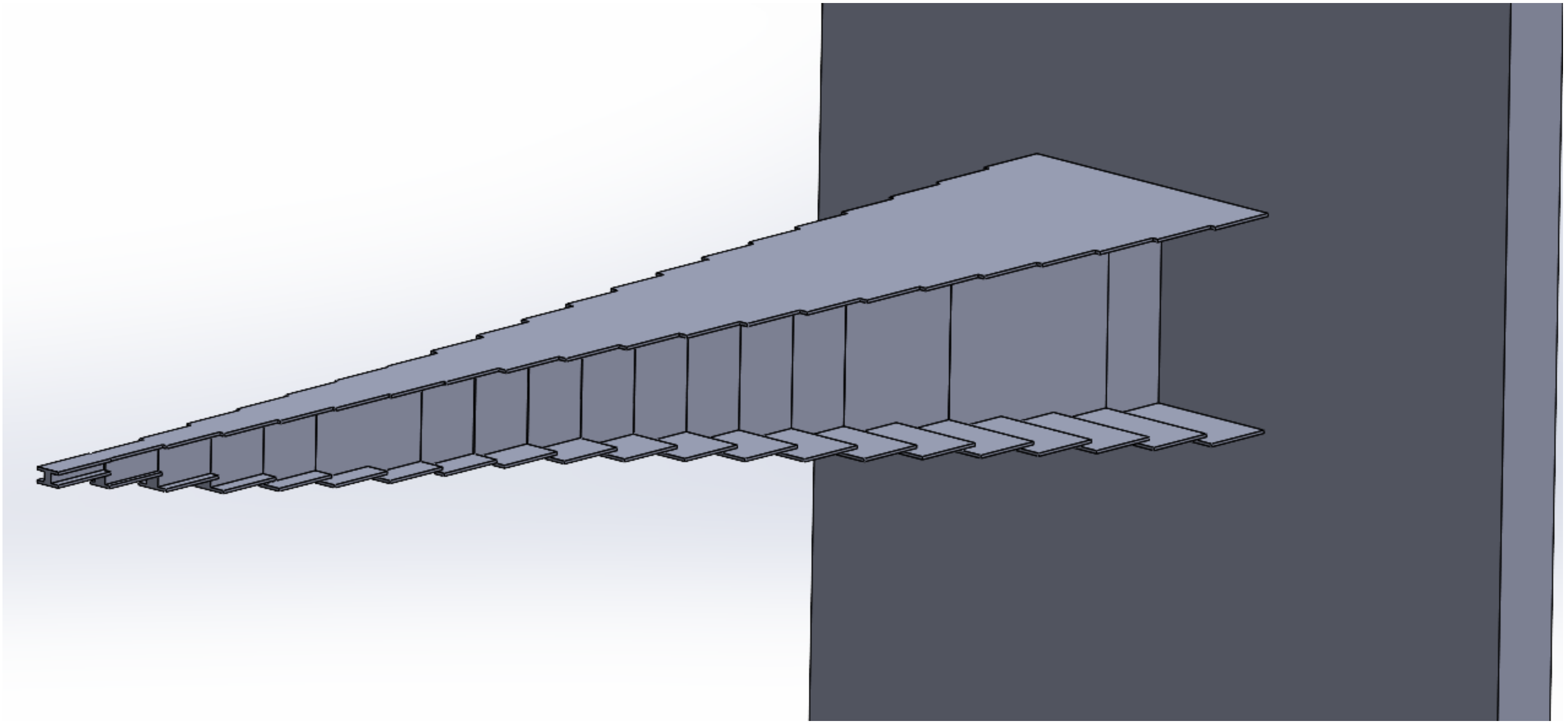
Weight: 13,340 lb

VARIED BEAM

Beam dimension table (x step = 1 in)

75	4.213181	0.500000	25.596260	34.590966	138.290925	5,608.494361	2
76	4.209297	0.500000	25.444324	34.342794	137.305823	5,501.937793	2
77	4.205350	0.500000	25.291292	34.093132	136.313406	5,396.479149	2
78	4.201328	0.500000	25.137227	33.842021	135.313850	5,292.120524	2
79	4.197218	0.500000	24.982192	33.589501	134.307333	5,188.864016	2
80	4.193007	0.500000	24.826248	33.335610	133.294030	5,086.711720	2
81	4.188683	0.500000	24.669460	33.080388	132.274117	4,985.665735	2
82	4.184233	0.500000	24.511888	32.823874	131.247772	4,885.728156	2
83	4.179644	0.500000	24.353597	32.566109	130.215171	4,786.901080	2
84	4.174904	0.500000	24.194649	32.307130	129.176490	4,689.186604	2
85	4.170000	0.500000	24.035105	32.046978	128.131905	4,592.586825	2
86	4.164920	0.500000	23.875030	31.785693	127.081593	4,497.103840	2
87	4.159649	0.500000	23.714485	31.523313	126.025731	4,402.739744	2
88	4.154177	0.500000	23.553534	31.259878	124.964495	4,309.496636	2
89	4.148490	0.500000	23.392239	30.995428	123.898060	4,217.376611	2
90	4.142575	0.500000	23.230662	30.730001	122.826605	4,126.381767	2
91	4.136420	0.500000	23.068867	30.463638	121.750305	4,036.514200	2
92	4.130012	0.500000	22.906915	30.196377	120.669336	3,947.776006	2
93	4.123338	0.500000	22.744870	29.928259	119.583876	3,860.169283	2
94	4.116386	0.500000	22.582794	29.659322	118.494100	3,773.696128	2
95	4.109143	0.500000	22.420751	29.389606	117.400185	3,688.358636	2
96	4.101596	0.500000	22.258801	29.119150	116.302308	3,604.158905	2
97	4.093733	0.500000	22.097009	28.847995	115.200644	3,521.099032	2
98	4.085541	0.500000	21.935437	28.576179	114.095370	3,439.181112	2
99	4.077007	0.500000	21.774147	28.303741	112.986663	3,358.407244	2
100	4.068118	0.500000	21.613202	28.030722	111.874700	3,278.779523	2
101	4.058863	0.500000	21.452665	27.757160	110.759656	3,200.300047	2
102	4.049227	0.500000	21.292599	27.483095	109.641708	3,122.970911	2
103	4.039199	0.500000	21.133065	27.208567	108.521032	3,046.794214	2
104	4.028765	0.500000	20.974128	26.933614	107.397806	2,971.772051	2
105	4.017914	0.500000	20.815848	26.658277	106.272204	2,897.906519	2
106	4.006632	0.500000	20.658290	26.382595	105.144405	2,825.199715	2
107	3.994906	0.500000	20.501516	26.106606	104.014583	2,753.653736	2
108	3.982725	0.500000	20.345588	25.830352	102.882917	2,683.270679	2
109	3.970074	0.500000	20.190568	25.553870	101.749581	2,614.052639	2
110	3.956942	0.500000	20.036521	25.277201	100.614753	2,546.001714	2
111	3.943316	0.500000	19.883508	25.000383	99.478609	2,479.120002	2
112	3.929184	0.500000	19.731592	24.723457	98.341325	2,413.409597	2
113	3.914531	0.500000	19.580835	24.446462	97.203079	2,348.872597	2
114	3.899347	0.500000	19.431301	24.169436	96.064045	2,285.511100	2
115	3.883617	0.500000	19.283052	23.892421	94.924401	2,223.327200	2

VARIED BEAM



This model serves as a visual approximation for the optimal varied beam size, with a step size of 12 in. Ideally, flanges of the beam would also connect.

THANK YOU

Aidan Crawford

EGR-228

12/11/2025